

PRATT, KS, USA

WMO: 745430

Lat: 37.702N

Lon: 98.747W

Elev: 595

StdP: 94.38

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 93978

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.8	-11.3	-19.4	0.7	-10.9	-17.1	0.9	-9.1	14.6	2.6	12.9	2.0	4.7	0	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	37.5	21.0	36.1	21.1	34.1	21.0	23.7	31.3	23.0	30.5	22.4	30.0	6.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	17.7	26.4	21.1	16.9	25.9	20.2	16.0	25.3	73.8	31.2	71.2	30.4	68.8	30.2	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.8	10.7	-18.5	39.6	3.4	1.8	-20.9	40.9	-22.9	41.9	-24.9	42.9	-27.4	44.2		
(5)			-18.9	24.8	3.2	0.9	-21.2	25.4	-23.0	26.0	-24.8	26.4	-27.1	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	0.8	1.8	8.2	12.7	18.5	24.5	26.5	25.2	21.5	14.1	7.1	1.3	(6)
(7)		DBStd	10.48	5.79	6.16	5.70	5.39	4.90	3.28	3.30	3.37	4.18	5.24	4.99	5.43	(7)
(8)		HDD10.0	1060	287	235	99	33	2	0	0	0	0	21	113	269	(8)
(9)		HDD18.3	2596	543	463	315	181	60	2	1	0	17	151	337	527	(9)
(10)		CDD10.0	2367	2	6	45	113	266	434	511	472	344	146	26	1	(10)
(11)		CDD18.3	859	0	0	2	11	65	186	253	214	111	18	0	0	(11)
(12)		CDH23.3	9878	1	2	33	177	712	2136	3057	2362	1159	226	15	0	(12)
(13)	CDH26.7	4744	0	0	4	53	263	1029	1649	1192	487	65	2	0	(13)	
(14)	Wind	WSAvg	5.0	4.8	5.0	5.6	5.9	5.1	4.3	4.0	4.7	5.0	5.0	4.9	(14)	
(15)	Precipitation	PrecAvg	711	17	27	48	65	99	102	88	90	53	55	30	26	(15)
(16)		PrecMax	963	55	74	171	145	246	211	167	201	143	172	91	90	(16)
(17)		PrecMin	420	0	0	0	5	22	29	6	9	14	3	1	0	(17)
(18)		PrecStd	134	14	23	41	36	52	48	43	54	31	43	29	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	22.2	27.2	32.1	35.0	38.7	40.0	39.9	37.0	31.5	25.9	17.4	(19)
MCWB			9.1	10.4	13.8	16.0	17.9	20.4	21.0	21.0	20.0	18.1	13.1	8.5	(20)	
2%		DB	15.9	18.4	23.1	27.6	32.1	35.9	37.7	37.3	33.9	28.0	22.2	14.9	(21)	
		MCWB	7.4	10.2	13.7	15.5	18.2	21.0	21.1	21.6	20.2	17.0	12.5	7.4	(22)	
5%		DB	12.5	15.0	21.1	25.1	29.8	33.9	36.3	35.1	32.0	26.1	18.1	12.3	(23)	
		MCWB	6.0	7.9	12.7	15.1	18.5	20.6	21.4	21.5	20.1	15.8	10.6	6.8	(24)	
10%		DB	10.0	12.0	17.8	22.4	27.4	32.3	34.2	32.8	29.7	22.8	16.0	10.0	(25)	
		MCWB	4.8	6.3	11.2	14.0	17.9	20.9	21.5	21.3	19.2	15.3	9.9	5.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	13.0	17.7	19.0	22.3	24.2	24.7	24.6	23.1	21.4	15.2	11.8	(27)
MCD B			16.1	16.5	22.7	25.4	27.5	31.6	31.8	32.8	29.4	27.0	20.2	13.6	(28)	
2%		WB	8.0	10.5	15.9	17.4	21.1	23.0	23.6	23.6	22.0	19.6	13.6	8.7	(29)	
		MCD B	14.7	17.1	20.3	23.2	27.3	31.0	31.2	31.5	28.5	24.6	19.2	13.4	(30)	
5%		WB	6.4	8.5	14.3	16.2	20.2	22.3	23.0	22.8	21.3	17.8	11.9	6.9	(31)	
		MCD B	12.3	15.2	18.8	22.2	26.5	30.1	30.6	30.5	27.8	23.1	17.2	11.7	(32)	
10%		WB	4.9	6.3	12.0	14.9	19.2	21.6	22.3	22.1	20.5	16.1	10.4	5.3	(33)	
		MCD B	9.5	11.4	17.3	21.4	25.4	29.1	30.2	29.8	27.0	21.6	15.3	9.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.2	12.2	13.3	13.3	12.7	12.9	12.8	12.5	12.8	13.4	13.2	11.5	(35)
MCDBR			17.1	18.3	17.4	17.5	15.6	15.1	15.6	15.6	14.9	17.1	17.5	15.5	(36)	
MCWBR			10.2	10.7	9.2	8.3	5.8	4.5	4.0	4.4	4.7	7.7	9.4	9.4	(37)	
5% WB		MCDBR	16.3	17.1	13.5	14.6	12.5	13.2	12.8	13.0	12.4	13.1	15.1	14.1	(38)	
		MCWBR	10.0	10.6	8.3	8.4	6.0	5.0	4.4	4.4	4.8	7.4	9.2	9.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.274	0.285	0.323	0.343	0.368	0.393	0.405	0.396	0.363	0.332	0.288	0.273	(40)	
taud		2.526	2.502	2.411	2.394	2.405	2.354	2.354	2.348	2.344	2.409	2.471	2.531	2.529	(41)	
Ebn at Noon		926	956	941	936	913	886	871	873	886	886	899	903	(42)		
Edh at Noon		76	89	108	116	117	124	124	121	107	91	75	71	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.60	3.38	4.47	5.56	6.36	7.08	6.93	6.18	5.14	3.79	2.84	2.29	(44)	
(45)		RadStd	0.24	0.42	0.33	0.32	0.48	0.46	0.37	0.34	0.32	0.43	0.29	0.20	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page